

PRAVIN KUMAR R

GENERATIVE AI & MACHINE LEARNING ENGINEER

Location: Kurinjipadi, Tamil Nadu – 607303 | Phone: +91 88259 38327 | Email: pravinrajavel@gmail.com

LinkedIn: linkedin.com/in/pravin-kumar-ds | GitHub: github.com/pravinkumardatafreak

PROFESSIONAL SUMMARY

Generative AI and Machine Learning Engineer specializing in autonomous multi-agent architectures, self-correcting RAG systems, and multimodal vision transformers. Proven track record designing stage-gated agent pipelines with sub-second retrieval latency, automated Critic LLM validation gates, and cost-observability waterfalls (\$0.0087/run). Proficient across Python, LangChain, PyTorch, Hugging Face, Groq API, Azure OpenAI, and production MLOps deployment.

SKILLS

Generative AI & LLMs: Multi-Agent Systems, Stage-Gated Orchestration, Self-Correction Loops, Critic Gates, LangChain, Groq, Multi-Agent RAG

NLP & Transformers: Hugging Face Transformers, DistilRoBERTa, Sentence-BERT (SBERT), TF-IDF, Cosine Retrieval, Tokenizer Optimization

Deep Learning & Vision: PyTorch, TensorFlow, Vision Transformers (ViT), Swin Transformer Vision, Shifted Window Attention, CNNs

Cloud, MLOps & Delivery: Azure (Azure OpenAI), AWS, GCP, Docker, CI/CD (GitHub Actions), Telemetry Waterfall, Streamlit, Git, Agile

PROJECTS

Unfoolable Lens: Stage-Gated Multi-Agent Rhetoric Deconstructor

Tools: Python, Multi-Agent Systems, GDELT API, Azure OpenAI

- Uncovered hidden editorial bias and rhetorical framing across **5 analytical perspectives** by architecting a 5-agent stage-gated NLP pipeline using Azure OpenAI.
- Delivered real-time news context grounding with **sub-second retrieval latency** by integrating live GDELT global news APIs and cross-referencing archival precedents.
- Achieved 100% operational transparency and minimized inference costs at **\$0.0087 per comprehensive run** by formulating a telemetry waterfall tracking token consumption and API expenditures.

Education Regulation Impact Analyzer (ERIA)

Tools: Groq LLaMA 3.3 70B, Multi-Agent RAG, DistilRoBERTa, LangChain, Hugging Face

- Cut compliance review cycles from days to **sub-15s latency** by orchestrating an autonomous 4-agent regulatory intelligence system on Groq LLaMA 3.3 70B with Multi-Agent RAG across **50+ page documents**.
- Eliminated hallucination risks and guaranteed **≥ 8.5/10 factual Grounding**, Consistency, and Completeness by engineering an automated Critic LLM evaluation gate with deterministic self-correction feedback loops.
- Empowered policy stakeholders with **7-class sentiment mapping** by integrating a local DistilRoBERTa transformer, deploying production web application on Hugging Face Spaces with automated PDF report generation.

Other Verified Projects: HuBMAP (0.6782 Dice, Swin Transformer Vision), IMDb 2-Stage SBERT Recommender, CryptoCast PyTorch Sequence Engine – on GitHub.

PROFESSIONAL EXPERIENCE

Data Science & Machine Learning Intern — Internship Studio

Jan 2026 – Feb 2026 | Remote

- Delivered **99.8% feature completeness** across **14,100+ session wearable records** by designing health analytics and data warehousing pipelines under 2-week Agile sprints.
- Reduced caloric prediction error by **89.2% over baseline models** (achieving an **R² of 0.9979** and **MAE of 3.78 kcal**) by developing a high-precision Random Forest regression model capturing nonlinear metabolic exertion.
- Cut executive reporting turnaround time by **75%** by deploying an interactive Streamlit and BI dashboard backed by K-Means and Principal Component Analysis (PCA) clustering users into 4 exertion zones.

EDUCATION

Master Data Science Program — HCL GUVI, IIT Madras Research Park Incubated

Completed Jun 2026

B.E. Civil Engineering (First Class Honors) — Rajalakshmi Engineering College, Anna University

Aug 2017 – May 2021

CERTIFICATIONS

Certifications: Data Science & Analytics (HCL GUVI), Build & Deploy AI Apps with Google AI Studio, ML for Earth Observation (EO College), Machine Learning & AI (HCL GUVI)

Additional: Stress Management (NPTEL)